

Lesson 4

Identification of modal parameters

1 Introduction

When modelling a mechanical system, the extraction of information from measurements performed on the system can be used to improve the model (a process known as “model updating”) and also to verify that the model reproduces to a good degree of accuracy the actual system’s behaviour, this latter procedure being referred to as “model validation”. Both model updating and validation require however that appropriate parameter identification techniques are used to extract valuable information from the available measurements. Particularly noteworthy to the topics treated in this course are the so-called *modal parameter identification techniques*, aimed at extracting some of the system’s modal parameters (e.g. natural frequencies, damping factors, modal shapes) from the measured system’s behaviour. Note that the identification of modal parameters can be of great use not only in the perspective of improving and/or validating a mathematical model, but can also be used directly (i.e. without the need to define a model) to analyse the dynamic behaviour of the system and, if needed, to identify changes in the system to improve its behaviour.

A simple example of modal parameter identification technique is the use of the “logarithmic decrement” to estimate the non-dimensional damping factor for a 1-d.o.f. system (cf. Cheli-Diana and Meirovitch). However, this method is limited in scope as it allows to identify one single modal parameter and the application to multi-d.o.f. systems can be problematic due to the difficulty of separating the different modal contributions in the free motion of the system.

In this lesson, we will introduce methods for the identification of modal parameters based on the measure of the system’s frequency response function (FRF). To this aim, we first derive in Section 2 an analytical expression for the FRF functions as function of the modal parameters (modal masses, damping, stiffness and modes of vibration). Then, we shortly describe how the experimental FRF functions for the same system can be derived from an experiment by measuring the input force applied on the system and the system’s response (Section 3). Finally, we introduce some simple methods to derive an estimate of the modal parameters out of the measurements performed, based on the analytical expression derived in Section 2. Different approaches to this latter task (i.e. to parameter identification in strict sense) are introduced. First, an approach capable of providing accurate results but requiring the use of a computer and of a suitable residual minimisation technique is presented in Section 4.1. Then a simpler approximation of the analytical expression of the FRFs is introduced, which is treated analytically to derive simple formulae for the identification of parameters such as the system’s natural frequencies, non-dimensional damping ratio, modes of vibration (Section 4.2).

2 Analytical expression of the FRF

We consider here a multi-degree of freedom lumped parameter system whose equations of motion are:

$$[m]\ddot{\underline{x}} + [c]\dot{\underline{x}} + [k]\underline{x} = \underline{f}(t) \quad (2.1)$$

and we assume the system is excited by a single input force f_k of harmonic nature, acting along the k -th independent coordinate of the system:

$$\underline{f}(t) = \begin{Bmatrix} 0 \\ \vdots \\ F_k \\ \vdots \\ 0 \end{Bmatrix} e^{i\Omega t} \quad (2.2)$$

Next, we apply the change of coordinates corresponding to the modal approach as introduced in the previous lesson:

$$\underline{x} = [\Phi] \underline{q} \quad (2.3)$$

with $[\Phi]$ the modal matrix (for the undamped system), so that the system's equations of motion in terms of the modal coordinates $\underline{q}$ are:

$$[\bar{m}] \ddot{\underline{x}} + [\bar{c}] \dot{\underline{x}} + [\bar{k}] \underline{x} = \bar{\underline{f}}(t) \quad (2.4)$$

with:

$$[\bar{m}] = [\Phi]^T [m] [\Phi]; [\bar{c}] = [\Phi]^T [c] [\Phi]; [\bar{k}] = [\Phi]^T [k] [\Phi]; \bar{\underline{f}} = [\Phi]^T \underline{f} \quad (2.5)$$

On account of the special expression (2.2) of the forcing vector, the vector $\bar{\underline{f}}$ of the modal forces takes the form:

$$\bar{\underline{f}}(t) = \begin{Bmatrix} x_{0k}^{(1)} F_k \\ \vdots \\ x_{0k}^{(i)} F_k \\ \vdots \\ x_{0k}^{(n)} F_k \end{Bmatrix} e^{i\Omega t} \quad (2.6)$$

We introduce here the additional assumption that by writing the equations of motion of the considered system according to the modal superposition principle, we obtain decoupled equations not only as far as the inertial and elastic terms are considered (which is always true, as shown in the previous lesson), but also for the viscous damping terms. Based on this additional assumption, the system's equations of motion written according to the modal superposition principle consist of n decoupled single-d.o.f. equations as follows:

$$\begin{cases} \bar{m}_{11} \ddot{q}_1 + \bar{c}_{11} \dot{q}_1 + \bar{k}_{11} q_1 = x_{0k}^{(1)} F_k e^{i\Omega t} \\ \dots \\ \bar{m}_{ii} \ddot{q}_i + \bar{c}_{ii} \dot{q}_i + \bar{k}_{ii} q_i = x_{0k}^{(i)} F_k e^{i\Omega t} \\ \dots \\ \bar{m}_{nn} \ddot{q}_n + \bar{c}_{nn} \dot{q}_n + \bar{k}_{nn} q_n = x_{0k}^{(n)} F_k e^{i\Omega t} \end{cases} \quad (2.7)$$

where $\bar{m}_{ii}$, $\bar{c}_{ii}$, $\bar{k}_{ii}$ are respectively the mass, damping and stiffness modal parameters for the i -th mode of vibration.

The solution in complex form of the single-d.o.f. equation of motion for the generic i -th modal coordinate reads:

$$q_i(t) = Q_i e^{i\Omega t} \quad (2.8)$$

with:

$$Q_i = \frac{x_{0k}^{(i)}}{-\Omega^2 \bar{m}_{ii} + i\Omega \bar{c}_{ii} + \bar{k}_{ii}} F_k \quad (2.9)$$

We now consider the steady-state response of the system evaluated at a given location and along a given component of displacement, denoted by $x_j(t)$:

$$x_j(t) = \sum_{i=1}^n x_{0j}^{(i)} q_i(t) \quad (2.10)$$

Introducing (2.8) and (2.9) in (2.10) we get:

$$x_j(t) = X_j e^{i\Omega t} \quad (2.11)$$

with:

$$X_j = \left(\sum_{i=1}^n \frac{x_{0j}^{(i)} x_{0k}^{(i)}}{-\Omega^2 \bar{m}_{ii} + i\Omega \bar{c}_{ii} + \bar{k}_{ii}} \right) F_k \quad (2.12)$$

and therefore the complex-valued FRF of the system evaluated at location j for a single unit input force applied at location k is:

$$G_{jk}(i\Omega) = \sum_{i=1}^n \frac{x_{0j}^{(i)} x_{0k}^{(i)}}{-\Omega^2 \bar{m}_{ii} + i\Omega \bar{c}_{ii} + \bar{k}_{ii}} \quad (2.13)$$

We note that (2.13) contains all the modal parameters (modal masses, damping, stiffness and modal shapes) for the first n modes of vibration of the system. Hence, we recognise the possibility to identify these modal parameters by fitting a set of experimental FRF values obtained from measurements using expression (2.13) or a simplified version of this expression, under the assumption that the set of modal parameters providing the best fit of the measured FRFs are the “actual” modal parameters of the system.

3 Extraction of the experimental FRFs from measurements

The experimental FRFs of the system can be derived from physical tests performed using a SIMO approach. To this aim, we apply a single excitation force $f_k(t)$ on the system and we derive the steady-state response of the system at multiple locations $x_j(t)$ $j=1, \dots, n_m$, with n_m the number of measured outputs for the system.

In this section we shortly recall how the FRFs $G_{jk}(i\Omega)$ can be extracted from n_m+1 measurements, i.e. the measure of the force applied at k (measured using a load cell) and the system's vibration measured at locations j , $j=1, \dots, n_m$. Note that for practical reasons, the response of the system is often measured using accelerometers instead of linear displacement transducers, in which case the response of the system in terms of displacements can be derived by double integration in time of the measured acceleration signals. This integration needs to be performed with particular care, to avoid drift effects produced by spurious low-frequency components in the measured signals, but this topic is not treated here and the reader is referred to classical books on mechanical measurements for more information in this regard.

We consider in this section two different cases of excitation: harmonic and impulsive excitation. Note that other types of excitation can be used in an experiment aiming at the extraction of the system's FRFs but, due to time constraints, only the two cases mentioned above are treated in this course. For information about other possible excitation cases, the reader is referred to (Cheli-Diana) and to the references cited in that book.

3.1 Harmonic excitation

We consider here the system excited by a single harmonic force at location k . Without loosing generality, we choose time $t=0$ corresponding to a positive maximum of the exciting force, hence the exciting force can be written in complex form as:

$$f_k(t) = F_k e^{i\Omega t} \quad (3.1)$$

with F_k the real-valued constant parameter representing the amplitude of the force (cf. Figure 3.1), and Ω the circular frequency of the force, which is related to the period T (also shown in Figure 3.1) by:

$$\Omega = \frac{2\pi}{T} \quad (3.2)$$

Assuming the system to behave linearly, the steady-state response of the system¹ evaluated at location j is also harmonic, with the same circular frequency Ω as the input force, and takes the form:

$$x_j(t) = X_j e^{i\Omega t} = |X_j| e^{i(\Omega t + \angle X_j)} \quad (3.3)$$

with X_j a complex-valued constant quantity defined in polar notation by its magnitude $|X_j|$ and phase $\angle X_j$. As shown in Figure 3.1, the magnitude $|X_j|$ represents the amplitude of the harmonic response, and the phase is related to the delay Δt of the first positive maximum of the response, evaluated with respect to the positive maximum of the input force occurring at $t=0$. From the time delay Δt the phase of the response can be evaluated as:

$$\angle X_j = -\frac{2\pi\Delta t}{T} = -\Delta t\Omega \quad (3.4)$$

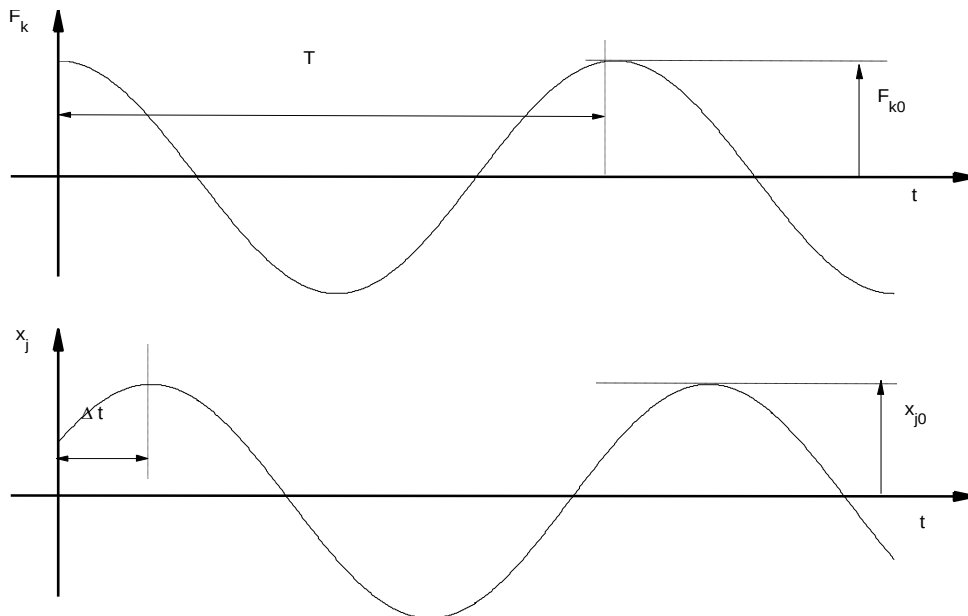


Figure 3.1: Input and output time histories for harmonic excitation of the system.

¹ i.e. the response evaluated after the transient generated by the application of the input force has decayed to negligible amplitude.

The experimental FRF is then derived as the ratio of x_{j0} over F_{k0} :

$$G_{jk}^{\text{exp}}(i\Omega) = \frac{|X_j|}{F_k} e^{i\angle X_j} \quad (3.5)$$

By changing in discrete steps the circular frequency Ω of the applied force, the experimental FRF of the system can be derived for a single input applied at k and multiple outputs evaluated at j with $j=1, \dots, n_m$. To simplify this time-consuming procedure, sometimes harmonic excitation is replaced by the so-called “swept-sine” excitation, which means the circular frequency of the exciting force is slowly changed with time to cover a pre-defined range of frequencies in one single test.

3.2 Impulsive excitation

Impulsive excitation is a practical approximation of the theoretical Dirac’s impulse. Since the Fourier transform of a theoretical impulse has constant magnitude at all frequencies, a ‘practical’ impulsive excitation is expected to apply more or less evenly excitation over some range of frequencies, allowing to derive the experimental FRF in this whole frequency range by means of a single test.

The usual way to produce impulsive excitation is to use an excitation hammer. This consists of a balanced hammer with a load cell fitted on its tip, which is used to measure the force generated by the hammer. Generally, the stiffness of the hammer tip impacting the system can be adjusted by choosing the among a set of tips made in different shapes and materials. This allows to control the frequency range in which excitation is applied.

Figure 3.2 shows the time histories of the exciting force $f_k(t)$ applied at the k -th coordinate of the system and of the system’s response evaluated at coordinate j . The exciting force is applied for a short time duration which depends on the stiffness of the tip of the hammer (a stiffer tip producing a shorter impact duration), and the waveform of the applied force is well approximated by a half-sine function padded with zero values. The response of the system contains different non-stationary harmonic contributions and, after the hammer loses contact with the system (i.e. no excitation is applied), corresponds to the free damped motion of the system, which in turn (for low damped systems) consists of the combination of a number of decaying oscillations.

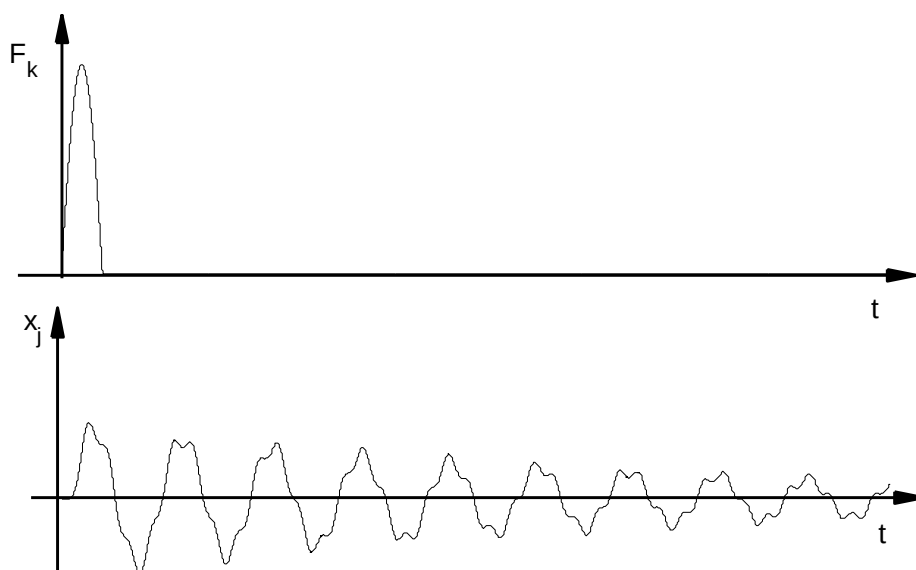


Figure 3.2: Input and output time histories for impulsive excitation of the system.

In order to extract the experimental FRF out of the measured time histories, a decomposition of the signal into its harmonic components shall be taken. Because of the nature of the signals, the direct use of the Fourier transform by analytical means is not possible. Therefore, the analog signals are first transformed into digital ones using analog to digital conversion (ADC) and then processed according to the “Discrete Fourier Transform” (DFT) algorithm, which provides a discrete spectrum for each signal. For the scopes of this lesson, we can consider the ADC as a sampling process, performed with sampling frequency f_s and duration of acquisition T . In this way, each signal (force at k , vibration at j , $j=1, \dots, n_m$) is transformed into a discrete set of n_s sampled values with:

$$n_s = Tf_s \quad (3.6)$$

The qualitative appearance of the discrete spectra for the force applied at k and the vibration measured at j is shown in Figure 3.3. For each measured quantity, the spectrum consists of two companion diagrams, one for the magnitude and one for the phase.

It is observed that:

- a.) the spectra are defined at discrete frequency values being the integer multiples of the “fundamental frequency” f_0 :

$$f_0 = \frac{1}{T} \quad (3.7)$$

note that the duration of the acquisition T is not a characteristic feature of the excitation applied (as was the case for the period T of the harmonic excitation treated in 3.1) and therefore can be chosen arbitrarily when performing the data acquisition process.

- b.) The discrete spectra are defined in a frequency range whose upper bound, according to the Shannon’s theorem, is one half of the sampling frequency f_s .
- c.) The spectrum of the impulsive force shows almost constant amplitude up to some maximum frequency which depends on the duration of the ‘real’ impulse and thus, ultimately, on the stiffness of the tip used. A stiffer tip will extend the pass band of the excitation, whereas a softer tip will concentrate the energy of the impact in a lower frequency range.
- d.) The amplitude of the harmonic components of the system’s response is otherwise changing with frequency, with larger amplitude values occurring at frequencies close to resonances of the system.

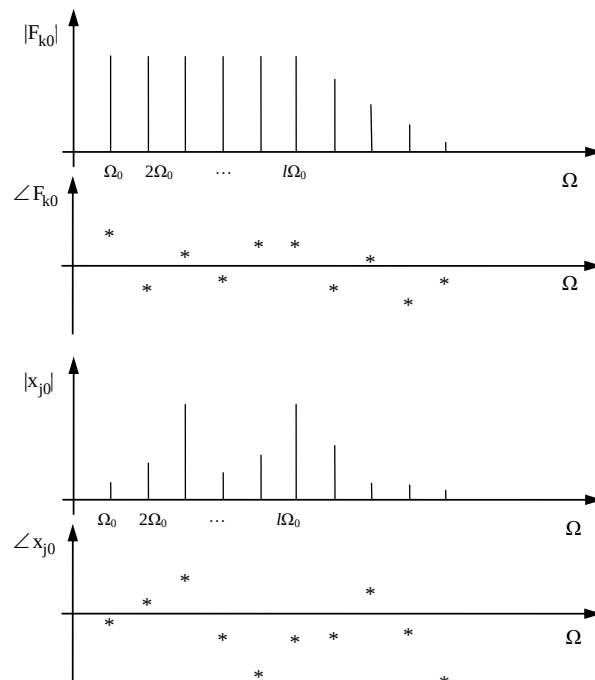


Figure 3.3: Discrete spectra of the input force and output vibration for impulsive excitation.

The experimental FRF is then defined at some discrete, equispaced values of the circular frequency, the same at which the DFT of the input and output signals are defined, i.e.:

$$\Omega_l = l\Omega_0 = l(2\pi f_0) = l\left(\frac{2\pi}{T}\right) \quad (3.8)$$

with l an integer value whose upper limit is discussed below. One possible way of defining the experimental FRF consists of taking the complex ratio of the harmonic component of the response $X_{j,n}$ over the corresponding harmonic component of the input force $F_{k,n}$:

$$G_{jk}^{\text{exp}}(i\Omega_l) = \frac{|X_{j,l}|}{|F_{k,l}|} e^{i(\angle X_{j,l} - \angle F_{k,l})} \quad (3.9)$$

The maximum value of index l that can be introduced in (3.9) is bounded by Shannon's theorem, i.e. $l f_0 < \frac{1}{2} f_s$ must be satisfied, so $l < \frac{1}{2} n_s$. Furthermore, the definition of the experimental FRF according to (3.9) shall not be used if the amplitude of the input force harmonic component is too low (note this might happen at frequencies much lower than the Shannon's limit, depending on the hammer tip used and on the sampling frequency), because that would imply a too low signal-to-noise ratio, which would lead to large errors in the definition of the experimental FRF.

4 Identification of modal parameters

In order to perform the identification of the system's modal parameters, we look for a set of modal parameters that produce the best fit of the analytical expression of the FRF (2.13) to the measured FRF $G_{jk}^{\text{exp}}(i\Omega)$ over a given frequency range.

This procedure is generally not easy to carry out in case the whole frequency range of interest is considered at once. Therefore, we observe that, based on (2.13), the FRF of the system is defined as the (complex valued) linear combination of n single-d.o.f. FRFs, one for each modal component considered.

If the system satisfies the following two conditions:

- i) the natural frequencies of the system are well separated;
- ii) the system is low damped

then in a relatively narrow frequency range centred on each resonance frequency of the system, the single term in (2.13) corresponding to the resonating modal component is prevailing over the other terms, as shown schematically in Figure 4.1

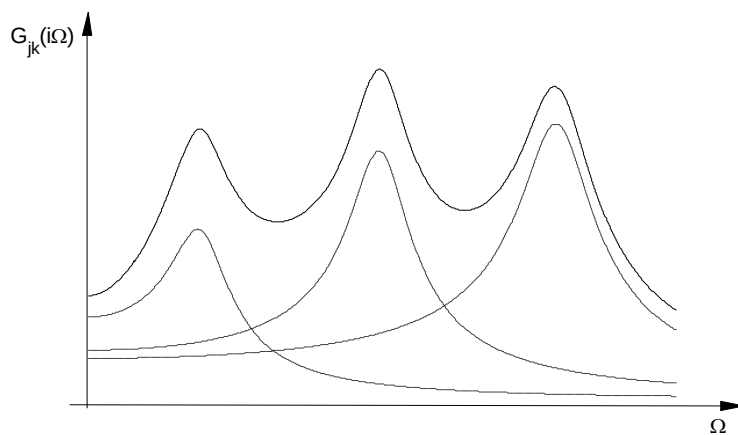


Figure 4.1: Qualitative graphic representation of the FRF modulus. In solid line: modulus of the overall FRF, in dashed line: modulus of the contribution from each single modal component.

This observation shows that the modal parameter identification process can be performed for one modal component at a time, by restricting the analysis to the frequency range in which the single modal component of interest becomes dominant in defining the system's FRF, so that the mathematical expression of the FRFs in this frequency range can be simplified substantially.

In the two following sub-sections, two different levels of simplification are introduced. In Section 4.1 a simplified expression still considering (at least in first approximation) the contribution of the non-resonating terms is introduced, whereas in 4.2 all terms in (2.13) are completely neglected, except the one excited in resonance. Clearly, the approach presented in 4.1 is likely to provide a more accurate identification of the system's modal parameters, as it involves a less severe simplification of the analytical FRF expression, however, the simplified FRF expression provided by this approach still requires to be treated numerically, using a residual minimisation technique, to perform the modal parameter identification, whereas the simpler approximation of the FRF introduced in 4.2 allows for an analytical treatment.

4.1 Modal parameter identification using residual minimisation

As anticipated above, we select a frequency range containing one single resonance frequency of the system, and we aim at identifying the parameters of the resonating modal component.

Thus, we approximate (2.13) as follows:

$$G_{jk}(i\Omega) = \underbrace{\frac{A_j + iB_j}{-\Omega^2 \bar{m}_{ii} + i\Omega \bar{c}_{ii} + \bar{k}_{ii}}}_{\text{resonating term}} + \underbrace{C_j + iD_j}_{\text{residual of modes resonating at higher frequencies}} + \underbrace{\frac{E_j + iF_j}{\Omega^2}}_{\text{residual of modes resonating at lower frequencies}} \quad (4.1)$$

The first term of this expression represents (with no approximation) the contribution to the FRF from the i -th modal component, which is here assumed to be the resonating one. The second term is a residual accounting in first approximation for the overall effect of all modal components resonating at frequencies higher than the selected frequency range, and the third term is a residual accounting for the effect of all modal components resonating at frequencies lower than the frequency range considered.

Each term is defined by a complex-valued “gain” parameter, namely $A_j + iB_j$ for the first term, $C_j + iD_j$ for the second term and $E_j + iF_j$ for the third term. The reason for considering complex-valued gains is that in case the damping in the system is of such nature that the damping matrix is not diagonalised by the modal matrix defined using the real-valued modes of vibration of the undamped system, it can be proved that a complex-valued modal superimposition can be defined using the complex-valued modes of vibration of the damped system, which leads to fully decoupled equations of motion for the system, also as far as the damping terms are concerned. Using this complex-valued modal superposition however, the ‘gain’ parameter $x_{0j}^{(i)} x_{0k}^{(i)}$ in (2.13) will be complex valued instead of real-valued.

To perform the identification of modal parameters for the resonating i -th mode, we prescribe:

$$\frac{A_j + iB_j}{-\Omega^2 \bar{m}_{ii} + i\Omega \bar{c}_{ii} + \bar{k}_{ii}} + C_j + iD_j + \frac{E_j + iF_j}{\Omega^2} = G_{jk}^{\text{exp}}(i\Omega) \quad (4.2)$$

for:

$$\begin{aligned} \Omega &= l\Omega_0 = l2\pi f_0 & l &= 1, \dots, n_f \\ j &= 1, \dots, n_m \end{aligned} \quad (4.3)$$

with n_f the number of frequency values at which the experimental FRF is available within the frequency range considered for parameter identification.

Using (4.2) and considering (4.3) we obtain a system of $n_f \times n_m$ equations in $3+6 \times n_m$ unknowns, the unknowns being:

$$\begin{aligned} \bar{m}_{ii}, \bar{c}_{ii}, \bar{k}_{ii} \\ A_j, B_j, C_j, D_j, E_j, F_j \quad j = 1, \dots, n_m \end{aligned} \quad (4.4)$$

If the number of frequencies n_f involved in this identification process is sufficiently large, an over-determined system of equations will be obtained. This can be solved using a least square approach, i.e. defining for each equation in the form (4.2) a (complex-valued) residual as the difference of the right hand side minus the left hand side, and looking for the set of modal parameters (4.4) that corresponds to a minimum of an overall residual function defined as the squares of the residuals summed over each equation.

Because the residual function depends non-linearly from the unknown modal parameters, this problem can only be solved using an iterative minimisation procedure. No detail is provided here about the numerical methods that can be used for this purpose, for this topic the reader is referred to any numerical analysis textbook.

Once solved this numerical problem, the modal mass, damping and stiffness parameters are directly obtained from the solution, cf. (4.4) whereas, based on comparison with (2.13), the real part A_j of the ‘gain’ of the resonating term represents the modal shape $x_{0j}^{(i)}$ of the resonating i -th mode evaluated at location j , multiplied by a common scaling factor $x_{0k}^{(i)}$ that can be removed by taking an appropriate normalisation of the mode. Of course, in order to rely on this result, it is required to check that:

- i) the imaginary part B_j of the ‘gain’ of the resonating term is small compared to A_j , so that the actual feasibility of using the real-valued modal matrix to perform the modal superposition is verified;
- ii) the real and imaginary parts of the ‘gains’ of the two residual terms in Eq. (4.1), i.e. coefficients C_j, D_j, E_j, F_j are all small compared to A_j .

if one of these checks is not satisfied (or both), this means that some of the assumptions underlying the modal parameter identification process are not satisfied, and hence the result of the identification shall be rejected or at least checked with care.

The process outlined above can be repeated for different frequency ranges, thereby leading to the identification of the modal parameters for all modal components of interest.

4.2 Modal parameter identification using analytical formulae

A more radical approximation of (2.13) consists of simply neglecting all terms except the resonating one (denoted below as the i -th term):

$$G_{jk}(i\Omega) = \frac{x_{0j}^{(i)} x_{0k}^{(i)}}{-\Omega^2 \bar{m}_{ii} + i\Omega \bar{c}_{ii} + \bar{k}_{ii}} \quad (4.5)$$

In this way, the FRF of the system is locally approximated as the FRF of a single-d.o.f. system. This allows to identify the following modal parameters of the system:

- un-damped natural frequency;
- non-dimensional damping ratio;
- mode of vibration.

using simple formulae, as show below.

Identification of the un-damped natural frequency

For a 1-d.o.f. system the absolute value of the FRF is maximum when the forcing frequency takes the value:

$$\Omega = \omega_i \sqrt{1 - 2h_i^2}$$

if the system is low damped, this value is very close to the un-damped natural frequency of the system, denoted here as ω_i . Hence, we identify here ω_i as the frequency at which $|G_{jk}(i\Omega)|$ is maximum within the considered frequency range.

Identification of the non-dimensional damping ratio – “phase derivative” method

For a 1-d.o.f. system, the phase of the FRF is approximately 0 at frequencies well below the resonance, $-\pi$ at frequencies well above the resonance and shows a transition from 0 to $-\pi$ around the resonance frequency; the lower the non-dimensional damping, the steepest the transition. Hence, measuring the “steepness” of the transition of the phase angle is a way to estimate the non-dimensional damping of the system. We take as a measure of this steepness the derivative of the phase angle with respect to the circular frequency Ω , evaluated at the resonant frequency $\Omega = \omega_i$ and we show below that this derivative is inversely proportional to the non-dimensional damping h_i of the resonating i -th modal component:

$$h_i = \frac{\bar{c}_{ii}}{c_{cr_i}} \quad (4.6)$$

To this aim, we observe that:

$$\angle G_{jk}(i\Omega) = -\text{artg}\left(\frac{\Omega \bar{c}_{ii}}{\bar{k}_{ii} - \Omega^2 \bar{m}_{ii}}\right) = -\text{artg}\left(\frac{2a_i h_i}{1 - a_i^2}\right) = -\text{artg}(b_i) \quad (4.7)$$

with:

$$a_i = \frac{\Omega}{\omega_i} \quad ; \quad b_i = -\frac{2a_i h_i}{1 - a_i^2} \quad (4.8)$$

Using the chain rule for derivation, we get:

$$\left. \frac{\partial \angle G_{jk}}{\partial \Omega} \right|_{\Omega=\omega_i} = \left. \frac{\partial \angle G_{jk}}{\partial a_i} \right|_{a_i=1} \frac{\partial a_i}{\partial \Omega} = \left. \frac{\partial \angle G_{jk}}{\partial a_i} \right|_{a_i=1} \frac{1}{\omega_i}$$

furthermore, also using the chain rule for derivation, we get:

$$\frac{\partial \angle G_{jk}}{\partial a_i} = \frac{\partial \angle G_{jk}}{\partial b_i} \frac{\partial b_i}{\partial a_i}$$

and, recalling that:

$$\frac{\partial \text{artg}(b_i)}{\partial b_i} = \frac{1}{1 + b_i^2} = \frac{(1 - a_i^2)^2}{(1 - a_i^2)^2 + 4a_i^2 h_i^2}$$

we get:

$$\frac{\partial \angle G_{jk}}{\partial a_i} = -\frac{(1-a_i^2)^2}{(1-a_i^2)^2 + 4a_i^2 h_i^2} \left[\frac{2h_i}{(1-a_i^2)} - \frac{2a_i h_i}{(1-a_i^2)^2} (-2a_i) \right] = -\frac{2h_i(1-a_i^2) + 4a_i^2 h_i}{(1-a_i^2)^2 + 4a_i^2 h_i^2}$$

Evaluating this derivative at $a_i=1$:

$$\left. \frac{\partial \angle G_{jk}}{\partial a_i} \right|_{a_i=1} = -\frac{1}{h_i}$$

so that, finally:

$$\left. \frac{\partial \angle G_{jk}}{\partial \Omega} \right|_{\Omega=\omega_i} = \left. \frac{\partial \angle G_{jk}}{\partial a_i} \right|_{a_i=1} \frac{1}{\omega_i} = -\frac{1}{h_i \omega_i} \quad (4.8)$$

This expression allows to estimate the damping factor h_i of the considered resonating modal component from the derivative of the FRF phase (for a single output j of choice). Clearly, repeating the estimation for different FRFs (i.e. for different outputs j) and averaging the result is likely to reduce the errors entailed by this simple estimation procedure.

Identification of the non-dimensional damping ratio – “half-power points” method

A second method which can be defined to estimate the non-dimensional damping factor h_i starting from (4.5) is the so-called “half power points” method. To introduce this method, we consider the plot of the magnitude of the non-dimensional FRF (i.e. the FRF divided by the value of the FRF at $\Omega=0$) vs. frequency for a single output j of choice, as shown in figure 4.2. On this plot, we define the so-called “quality factor” Q as the maximum amplitude of the FRF. Next, we identify the two frequencies ω_1 and ω_2 at which the modulus of the FRF is $Q \frac{\sqrt{2}}{2}$. These points are called the “half-power points”, because often in engineering problems the power of a given quantity is related with the squared value, so that if the amplitude drops down from the maximum value by a factor $\frac{\sqrt{2}}{2}$, then the “power” (in some sense) is decreased by a factor $\frac{1}{2}$.

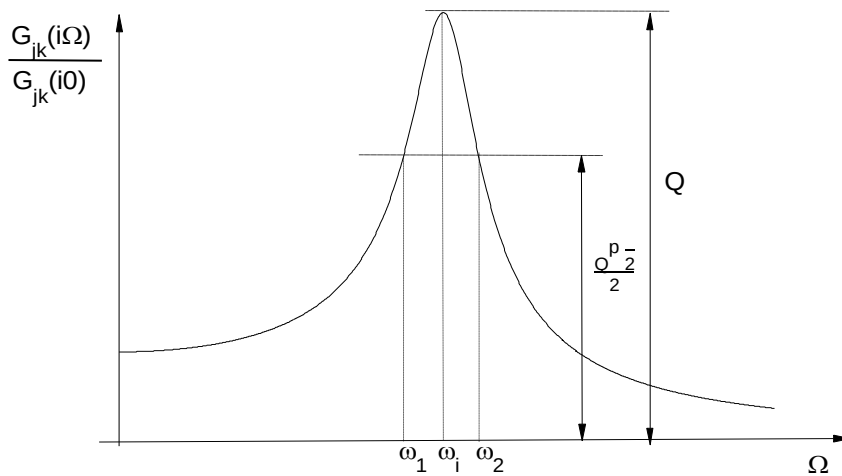


Figure 4.2: Half-power points for a 1-d.o.f. frequency response function.

Assuming the non-dimensional damping $h_i \ll 1$, it is possible to show by trivial mathematics (not reported here for the sake of brevity, the interested reader is referred to either Cheli-Diana or Meirovitch for a proof) that:

$$h_i = \frac{\omega_2^2 - \omega_1^2}{4\omega_i^2} \quad (4.9)$$

providing another simple formula that can be used, besides (4.8), to identify the non-dimensional damping factor of the resonating modal component in the frequency range considered.

Identification of the mode of vibration

Finally, we provide a simple formula that can be used to identify the shape of the mode of vibration evaluated at the measuring locations $j, j=1, \dots, n_m$.

To this aim, we evaluate (4.5) for $\Omega = \omega_i$.

Because:

$$\omega_i = \sqrt{\frac{\bar{k}_{ii}}{\bar{m}_{ii}}} \quad ; \quad \bar{k}_{ii} = \omega_i^2 \bar{m}_{ii}$$

at this special frequency value we get:

$$G_{jk}(i\Omega)\big|_{\Omega=\omega_i} = \frac{x_{0j}^{(i)} x_{0k}^{(i)}}{i\Omega \bar{c}_{ii}} = -i \frac{x_{0k}^{(i)}}{\Omega \bar{c}_{ii}} x_{0j}^{(i)} \quad (4.10)$$

this equation shows that the shape of the i -th resonating mode of vibration evaluated at location j is equal to the imaginary part of the frequency response function G_{jk} evaluated at the resonance frequency ω_i , multiplied by a common scaling factor $-\frac{x_{0k}^{(i)}}{\Omega \bar{c}_{ii}}$. This scaling factor can be removed by

taking an appropriate normalisation of the mode.

We note that the resonance frequency ω_i was previously identified and, hence, it is here considered as a known parameter. Also, we note that, based on (4.10), one might be tempted to take the magnitude of the FRF G_{jk} at the resonance frequency as an equivalent estimate of the modal shape at measuring point j . We recall however that (4.10) is derived from (4.5), which in turn comes from (2.13) neglecting all the non-resonant terms. Because in the frequency range considered the non-resonant terms are excited either in the quasi-static or seismographic region, their contribution is approximately real valued (in other words, the phase associated with these non-resonant terms is either close to zero or close to $-\pi$). Hence, taking the imaginary part of the FRF instead of the magnitude provides a better rejection of the effect of the non-resonating modes.